

**Amaan Choughale****(Attempt 1 of 1)****Test Information****QUESTION 1: CALCULATED NUMERIC****10****out of 10 points**

The generating function for a binary nonideal liquid mixture is

given by the equation $\frac{G^E}{RT} = -1.5 x_1 (1 - x_1)$. What is the

value of the liquid activity coefficient of the first component at infinite dilution. Round your numerical answer to **three decimal digits**



Given Answer: .223

Correct Answer: 0.223

Answer range +/- 0.001 (0.222 - 0.224)

QUESTION 2: CALCULATED NUMERIC**10****out of 10 points**

The generating function for a binary **EQUIMOLAR** nonideal liquid

mixture is given by the equation $\frac{G^E}{RT} = -1.5 x_1 (1 - x_1)$. What

is the value of the fugacity of the **first** component in the mixture. Assume that the vapor phase behaves as an IDEAL GAS mixture and that the Poynting factor is equal to one. Express your result in **kPa** and round your numerical answer to **a whole number**.



Data: $P_1^{sat} = 7.7 \text{ bar}$ and $P_2^{sat} = 3.9 \text{ bar}$

Given Answer: 265

Correct Answer: 265

Answer range +/- 5 (260 - 270)

QUESTION 3: TRUE/FALSE

10

out of 10 points

The fugacity coefficient of a component inside an IDEAL GAS mixture is equal to one.



Given Answer: True

Correct Answer: True

QUESTION 4: TRUE/FALSE

10

out of 10 points

The fugacity coefficient of a component inside an IDEAL LIQUID mixture is equal to one.



Given Answer: False

Correct Answer: False

QUESTION 5: MULTIPLE CHOICE

10

out of 10 points

Consider an IDEAL LIQUID mixture in equilibrium with NON-IDEAL GAS mixture. Which of the following expressions for the K-value of the component 1 is the correct one



Given Answer:



$$K_1 = \frac{f_1^L}{P \hat{\phi}_1^V}$$

Correct Answer:



$$K_1 = \frac{f_1^L}{P \hat{\phi}_1^V}$$

QUESTION 6: CALCULATED NUMERIC

0

out of 10 points

Calculate the numerical value of the thermodynamic expression



$\frac{1}{RT} \left. \frac{\partial \rho U}{\partial \rho} \right|_S$ for ammonia at T=45 °C and P=20 bar.

Given Answer: 2.56

Correct Answer: -24.6

Answer range +/- 0.1 (-24.7 - -24.5)

QUESTION 7: CALCULATED NUMERIC

15

out of 15 points

Consider a NON-IDEAL Liquid mixture of Methyl tert-butyl ether ($C_5H_{12}O$: Component **1**) and Water (H_2O : Component **2**). The Vapor Liquid Equilibrium behavior of the mixture is described by the **NRTL** method of liquid activity coefficients with binary interaction parameters $A_{12}=A_{21}=0.0$ $B_{12}=686.1436$

$B_{21}=1106.9764$ and $C=0.3$. The mixture exhibits an azeotrope.



Calculate the temperature of this azeotrope at $P=5$ bar. Assume that the vapor phase behaves as an IDEAL GAS mixture. Express your result in **deg C** and round your numerical answer to a **single decimal digit**.

Hint: You may start with an initial guess for the azeotrope temperature $T^{(0)} = 100\text{ }^{\circ}C$

Given Answer: 104.6

Correct Answer: 104.6

Answer range +/- 0.1 (104.5 - 104.7)

QUESTION 8: CALCULATED NUMERIC

10

out of 10 points

Consider a NON-IDEAL **Equimolar** Liquid mixture of Methyl tert-butyl ether ($C_5H_{12}O$: Component **1**) and Water (H_2O :

Component **2**). The Vapor Liquid Equilibrium behavior of the mixture is described by the **NRTL** method of liquid activity coefficients with binary interaction parameters $A_{12}=A_{21}=0.0$

$B_{12}=686.1436$ $B_{21}=1106.9764$ and $C=0.3$. The mixture is



maintained at $120\text{ }^{\circ}C$. Calculate the bubble point pressure. Assume that the vapor phase behaves as an IDEAL GAS mixture. Express your result in **bar** and round your numerical answer to a **single decimal digit**.

Given Answer: 7.4

Correct Answer: 7.4

Answer range +/- 0.1 (7.3 - 7.5)

QUESTION 9: CALCULATED NUMERIC

15

out of 15 points

A **liquid** mixture of Propane (Component 1) and Ethane (Component 2) with overall molar composition $[z_1, z_2] = [0.90, 0.1]$ at $T_1 = 40^\circ\text{C}$ and $P_1 = 20$ bar, is heated isobarically to $T_2 = 50^\circ\text{C}$. At the end of the heating process the mixture becomes two-phase. Calculate the amount of heat required for the heating process if the molar flow rate of the mixture is 100 kmol/hr. Assume that both phases of the mixture follow the Peng-Robinson Equation of State with a binary interaction parameter $k_{12} = 0.0011$. Express your result in **kW** and round your numerical answer to **a whole number**.



Given Answer:  122

Correct Answer:  124

Answer range +/- 4 (120 - 128)